CSD-380: DevOps
Module 1: The History of DevOps
Assignment 3: The History of DevOps
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The History of DevOps

The Lean Movement

Lean is a scientific approach to management attempting to maximize value created and reduce waste, thereby expanding the enterprise's production possibilities frontier and its profit, and create value and goodwill in the customer. The history of Lean is inseparable from industrialization. Historically just about every development in automation, interchangeable parts, assembly-line construction, observation-based process improvement, etc has advanced the cause of Lean manufacturing.

Lean is receptive to observations and feedback from all stakeholders, including regular assembly line workers. Process improvement suggestions are evaluated on a kind of scientific basis, and if found to be an objective improvement, the new "best practice" is adopted fully and uniformly.

There does not appear to be a history of Lean out there that does not mention Henry Ford. I will note that Ford was famous for brutal, top-down micro-optimization of his line workers, often resulting in repetitive strain injuries from the prescribed, optimized motions. Ford was also a pioneer of the factory town, and many research papers have been written on the dangerous reach and scope of his paternalism. But he also changed manufacturing in the U.S., using interchangeable parts, assembly lines with automated conveyors, and a methodical approach to continuous process improvement.

Another frequently-cited point in the timeline is the Toyota Production System. Their concept of "jidoka" was originally motivated by threads breaking in automated looms. When a thread breaks, and the machine continues, the workpiece is often ruined. Jidoka is kind of like automation with a human touch - add a button for a machine operator to stop the machine when a defect appears, and you get the ability to maybe mend the weave or salvage as much cloth as possible.

The inventor of the "jidoka" method had a son, who went on to found the Toyota Motor Corporation. They carried this concept forward into their own car production, along with inspiration from Ford about flow-based production. So the Toyota Production System combined sizing machine capacity to deal with the actual flow of material through the plant, with measures to prevent and remedy defects throughout the process, preferably automated.

One of the designers of the Toyota Production System, Taichi Ohno, went on to create Kanban. He realized that products leave store shelves at whatever rate customers select, which can vary. Manufacturing at this ever-changing rate requires careful planning, and so Kanban is a kind of physical, card-based storyboarding system to track and follow product runs as they make their way through the manufacturing pipeline. This system is virtually identical to an agile storyboard.

Many of these process improvements, indeed the very attitude of continuous process improvement, translate very elegantly to software development, and so, for example, an agile storyboard today is strikingly similar to the physical cards used in Kanban, and the careful attention to software fault detection and fail-fast design can be traced directly back to TPS.

The Agile Manifesto

The document is an actual "manifesto" written by 17 software development experts and executives. To really understand it you have to understand what came before: Waterfall.

When you take a software contract, someone first sends you a Request For Proposal. You do a requirements analysis on this proposal, form a comprehensive plan for what you will offer, and then write a proposal. The client accepts your proposal and then you start another round of specific subsystem designs which must conform to the overall integrated system design. Once the design is complete, you start developing. Hopefully you have designed a modular system that you can fire at individual, autonomous teams that can work efficiently in parallel, knowing that they need only meet the software specifications they've been given. Once every piece is complete, the solution is integrated, problems found and fixed, and the software is delivered. Now you get paid. You may have to perform minor maintenance on the system depending on your contract. You will get paid more for this. Job complete.

This attitude towards the software process, for one, didn't leave much room for improvement. The process calcifies, specialized teams become siloed, and people develop "throw it over the wall" mentalities where defects are someone else's problem to deal with. You also had this kind of cargo culting going on with firms like Six Sigma and Arthur Andersen. Companies such as these promoted themselves as collecting industry best-practices, but prescribed slow, rigid, unwieldy, document-heavy processes that had little room to evolve and improve over time.

The Agile Manifesto authors thought we could do better, not just by improving any of the linear operations, but by rethinking the shape of development to be more responsive to changing customer needs rather than meticulous ahead-of-time planning. Agile stresses a close, collaborative relationship with the customer, focusing on delivering them high-quality, bug-free software, even if documentation has to take a back seat.

The goal here was eliminating waste, Toyota-style, and creating more sustainable development, where the central goal is working software - actual value delivered to the customer.

The Continuous Delivery Movement

Continuous Delivery builds on Lean and Agile to establish a kind of automated conveyor for Agile's continuous stream of updates. Agile devs work collaboratively with their clients and create changes that satisfy their changing needs. Continuous delivery makes one small tweak to that process: It makes every change releasable on its own.

So every time a story card is completed, the continuous delivery "conveyor belt" can pull it along through a series of automated tests, integrating it and verifying its safety and correctness. These systems often push the completed and verified builds out to a test environment for final, human testing and approval. These final approvals rarely take long, and so the efficiency of such systems is incredibly high. And yet, this creates a kind of "jidoku", where at a very granular level a worker can still hit the "emergency stop button" and fix problems before they get out of hand. Instead of preventing material damage, this prevents system downtime, but it's the same concept.

All of these processes build on each other, and we're certainly not done improving. I'm certain we'll see at least one more process revolution in our lifetimes. It's just the natural progression of the Lean mindset.

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